Water Containers Moves

Time limit: 1.00 s **Memory limit:** 512 MB

There are two water containers: container A has volume a and container B has volume b. Your want to measure x units of water using the containers.

Initially both containers are empty. On each move, you can fill a container, empty a container or move water from a container to another container. When you move water, you must always fill or empty at least one container. After the moves, container A must have x units of water.

Find a sequence of moves where the total amount of moved water is minimum or state that it is impossible to measure the water.

Input

The only line has three integers a, b and x.

Output

First print two integers n and m: the number of moves and the total amount of water moved. After that print a sequence of n moves. Each move must move at least one unit of water and be one of the following:

- `FILL A`: fill container A
- `FILL B`: fill container B
- `EMPTY A`: empty container A
- `EMPTY B`: empty container B
- `MOVE A B`: move water from container A to container B
- `MOVE B A`: move water from container B to container A

If it is not possible to measure the water, only print -1.

Constraints

- $1 \leq a, b, x \leq 1000$

Example

Input	Output

`5 3 4`	`6 19`
	`FILL A`
	`MOVE A B`
	`EMPTY B`
	`MOVE A B`
	`FILL A`
	`MOVE A B`